



CHARUSAT
CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

Faculty of Technology & Engineering
Bachelor of Technology Programme
(B.Tech. CE/CSE/IT)
ACADEMIC YEAR 2026-27

ACADEMIC REGULATIONS & SYLLABUS

(Choice Based Credit System)



Faculty of Technology and Engineering
B. Tech. Programme
Chandubhai S. Patel Institute of Technology
Devang Patel Institute of Advance Technology and Research





CHARUSAT

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

Academic regulations recommendations are provided to ensure uniform system of education, programmes duration, eligibility criteria for admission, course credits distribution, teaching and examination pedagogy, detailed syllabus with reference material.

1) System of Education

The Charotar University of Science and Technology (CHARUSAT) shall follow Choice based Credit System (CBCS) with Semester pattern at Undergraduate and Master levels. Each semester will be at least of 90 working days. Apart from the programme core courses, provision for choosing University level electives and Programme electives are available under the CBCS.

2) Duration of Programme

i) Undergraduate programme	(B.Tech.)
Minimum	8 semesters (4 academic years)
Maximum	14 semesters (7 academic years)

3) Eligibility for admissions

As enacted by Government of Gujarat/AICTE/UGC from time to time.

4) Mode of admissions

As enacted by Government of Gujarat from time to time.

5) Programme Structure and Credits

As per annexure – 1 attached

6) Attendance

- 6.1. Students are expected to maintain 100% attendance in all courses. However, students may involuntarily have to miss classes due to illness or some family emergency; students are permitted to maintain a minimum attendance of 75% with producing

proof or reason for the absence. In case of medical exigencies, the student/parent should inform the principal immediately through call or by email. Within a week, starting from the day of absence, the proof of medical exigency must be submitted to the Principal's office.

- 6.2. Unauthorized absence will be considered as part of the discretionary 25% for fulfilling the minimum 75% attendance requirement for appearing in the examination.
- 6.3. Students nominated/sponsored by the University to represent in various forums like seminars/conferences/workshops/competitions or taking part in co-curricular/extra-curricular events will be given attendance credit provided the student applies in writing for such a leave in advance and obtains sanction from the Principal of his/her Institute for academic related requests.

7) Course Evaluation

7.1. The performance of every student in each course will be evaluated as follows:

- 7.1.1 Internal evaluation by the course faculty member(s) based on continuous assessment. The respective department /institute will conduct the continuous assessment. The course faculty members shall share the pedagogy related to the continuous evaluation with the students.
- 7.1.2 Final end-semester examination shall be conducted by the University through written paper, practical test, oral test, presentation by the student or a combination of these.
- 7.1.3 The weightages of continuous assessment and end-semester university examination in overall assessment shall depend on individual course as approved by Academic Council through Faculty Board and Board of Studies.
- 7.1.4 The performance of candidate in continuous assessment and in end-semester examination together (if applicable) shall be considered for deciding the final grade in a course.
- 7.1.5 In order to earn the credit in a course a student has to obtain grade other than FF.

7.2. Performance in continuous assessment and end-semester University Examination

- 7.2.1 Minimum performance with respect to continuous assessment as well as end-semester university examination will be an important consideration for passing a course.

- 7.2.2 If a candidate fails to obtain minimum required overall percentage of marks (36%), student has to repeat the examination till the minimum required overall percentage obtained.

8) Grade Point System

- The total of the internal evaluation marks and end semester examination marks in each course will be converted to a letter grade on a ten-point scale as per the following scheme:
- Proposed Grading Scheme to awarding letter grade and grade point as per NEP 2020

Letter Grade	Grade Point	Grading Scheme for Mark (In %)
O (Outstanding)	10	96.0-100
A+ (Excellent)	9	86.0-95.9
A (Very Good)	8	76.0-85.9
B+ (Good)	7	66.0-75.9
B (Above Average)	6	56.0- 65.9
C (Average)	5	46.0 – 55.9
P (Pass)	4	36.0 – 45.9
F (Fail)	0	Below 36.0
Ab (Absent)	0	Absent

➤ The minimum passing marks for each pattern of evaluation are 36%

- The student's performance in any semester will be assessed by the Semester Grade Point Average (SGPA). Similarly, his/her performance at the end of two or more consecutive semesters will be denoted by the Cumulative Grade Point Average (CGPA). The SGPA and CGPA are calculated as follows:

(i)
$$SGPA = \frac{\sum C_i G_i}{\sum C_i}$$
 where C_i is the number of credits of course i

G_i is the Grade Point for the course i

and $i = 1$ to n , n = number of courses in the semester

(ii) $CGPA = \frac{\sum C_i G_i}{\sum C_i}$ where C_i is the number of credits of course i

G_i is the Grade Point for the course i

and $i = 1$ to n , n = number of courses of all semesters up to which CGPA is computed.

9) Award of Class

- ❖ The class awarded to a student in the programme is decided by the final CGPA as per the following scheme:

Award of Class	CGPA Range
First Class with Distinction	$CGPA \geq 7.0 \text{ \& } \leq 10.0$
First class	$CGPA \geq 6.0 \text{ \& } < 7.0$
Second Class	$CGPA \geq 5.0 \text{ \& } < 6.0$
Pass Class	$CGPA < 5.0$

10) Detention Criteria

- ❖ A student will be promoted to next year only if he/she has cleared all the courses of the year he/she is studying in.

Link: <https://charusat.ac.in/> => Student's Corner => Detention Rules



CHARUSAT
CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

**CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
(CHARUSAT)**

**FACULTY OF TECHNOLOGY & ENGINEERING
(FTE)**

CHOICE BASED CREDIT SYSTEM

FOR

**BACHELOR OF TECHNOLOGY & ENGINEERING
(B.Tech. CE/CSE/IT)**

Choice Based Credit System

With the aim of incorporating the various guidelines initiated by the University Grants Commission (UGC) to bring equality, efficiency and excellence in the Higher Education System, Choice Based Credit System (CBCS) has been adopted. CBCS offers wide range of choices to students in all semesters to choose the courses based on their aptitude and career objectives. It accelerates the teaching-learning process and provides flexibility to students to opt for the courses of their choice and / or undergo additional courses to strengthen their Knowledge, Skills and Attitude.

1. CBCS – Conceptual Definitions / Key Terms (Terminologies)

Types of Courses: The Programme Structure consist Foundation courses, Core courses, Elective courses, Non-credit (audit) courses and SWAYAM MOOCs.

1.1 Foundation Course

These courses are offered by the institute in order to prepare students for studying courses to be offered at higher levels.

1.2 Core Courses

A Course which shall compulsorily be studied by a candidate to complete the requirements of a degree / diploma in a said programme of study is defined as a core course. Following core courses are incorporated in CBCS structure:

A. University Core courses(UC):

University core courses are compulsory courses which are offered across university and must be completed in order to meet the requirements of programme. Environmental science will be a compulsory University core for all Undergraduate Programmes.

B. Programme Core courses(PC):

Programme core courses are compulsory courses offered by respective programme owners, which must be completed in order to meet the requirements of programme.

1.3 Elective Courses

Generally, a course which can be chosen from a pool of courses and which may be very specific or specialised or advanced or supportive to the discipline of study or which provides an extended scope or which enables an exposure to some other discipline / domain or nurtures the candidates proficiency / skill is called an elective course. Following elective courses are incorporated in CBCS structure:

A. University Elective Courses(UE):

The pool of elective courses offered across all faculties / programmes. As a general guideline, Programme should incorporate 2 University Electives of 2 credits each (total 4 credits).

B. Programme Elective Courses(PE):

The programme specific pool of elective courses offered by respective programme.

1.4 Non Credit Course (NC) - AUDIT Course

A 'Non Credit Course' is a course where students will receive Participation or Course Completion certificate. This will be reflected in Student's Grade Sheet but the grade of the course will not be considered to calculate SGPA and CGPA. Attendance and Course Assessment is compulsory for Non Credit Courses.

1.5 Credit Transfer through SWAYAM MOOCs

CHARUSAT provides credit transfer as per UGC guidelines to all the students from SWAYAM against elective courses. The credit transfer is offered in two modes: (a) Partial credit transfer (b) Full credit transfer.

1.6 Medium of Instruction

The Medium of Instruction will be English.

In consonance with the National Education Policy (NEP) 2020 and the guidelines of the University Grants Commission (UGC), Charotar University of Science and Technology (CHARUSAT) implements the Multiple Exit scheme in their Bachelor of Technology programme.

Facilitating multiple exit options with UG certificate/UG diploma/or UG degree depending upon the number of credits secured. Skill component with progressive enhancement in skills in respective disciplines is introduced in the curriculum right from the 1st year of the program to ensure the student employability after every exit.

Students may be permitted to take a break from the study during the period of study but the total duration for completing the programme shall not exceed 7 years. If student wishes, he/she shall be allowed to re-enter the degree programme within three years and complete the degree programme within the stipulated maximum period of seven years. Table 1 shows the exit qualifications along with credit requirements.

Table 1 Exit Qualifications along with Credit Requirements

ACADEMIC LEVEL	EXIT QUALIFICATION AND CREDITS REQUIRED	NATIONAL CREDIT LEVEL (NCrF)
1st year of UG Degree	UG Certificate will be awarded Minimum 40 credits followed by an exit 4-credit skills enhancement course	4.5
2nd year of UG Degree	UG Diploma will be awarded Minimum of 80 credits followed by an exit 4-credit skills enhancement course	5.0
3rd year of UG Degree	B.Sc. will be awarded Minimum of 120 credits	5.5
4th year of UG Degree	B. Tech. will be awarded Minimum of 160 credits, with minimum of 40 credits each at level	6.0

Guidelines for multiple exit along with awarding UG Certificate, UG Diploma, and Degrees:

A. Exit after First Year

Award: UG Certificate in XXXX

Student must undertake a 4-Credits skill enhancement course before commencement of 2nd year. This shall require 120 hours engagement in the relevant industry/organization. Student shall have an option to continue the 2nd year onward study.

The student shall be awarded with “UG Certificate in XXXX”, with redemption of credits from ABC. Total credits redemption shall be 1st year credits + 4 credits earned in summer.

B. Exit after Second Year

Award: UG Diploma in XXXX

Student must undertake a 4-Credits summer internship/ skill enhancement course before commencement of 3rd year. This shall require 120 hours engagement in the relevant industry/organization. Student shall have an option to continue the 3rd year onward study.

The student shall be awarded with “UG Diploma in XXXX”, with redemption of credits from ABC. Total credits redemption shall be 1st and 2nd year credits + 4 credits earned in summer.

C. Exit after Third Year

Award: B.Sc. Degree in XXXX

The student shall be awarded with “B.Sc. Degree in XXXX”, with redemption of 3 years’ credits from ABC.

4-Credit (120 hrs.) Skills Enhancement Courses:

Level/ Branch	CE	IT	CSE	AIML
1st year of UG Degree	CEUS101 : Web Development	ITUS102: Introduction of Web Technologies	CSUS101:Data Analysis	AIUS101 : Data Analysis

Value Added Courses:

Inclusion of 2 credits courses on Community Service/ NSS/NCC/ Sports; and provision to earn extra credits based on undertaking Research/ Academic/Cultural/ and other Developmental activities is introduced.

- The component of ‘Value-added Courses’ could be enriched to include alternatives that could contribute to the holistic development of the students. In light of this, it is proposed to include a 02 credit Course on Community Service/ NSS/NCC/Sports as a compulsory course in the existing curricula offered across all UG programs at CHARUSAT. It is mandatory for all UG students (2024-25 admission batch) to undertake this course and the course shall not account for the overall CGPA.
- Provision to earn extra credits (**Skill Augmentation** course) based on co-curricular & extra-curricular activities is proposed. It was emphasized that the extra credits could help in fuller realization of the Graduate Attributes laid down by the university as well as could serve as an important ingredient facilitating the progression and recognition of fast learners. It is mandatory for all UG students (2024-25 admission batch) to undertake this course during the study and the credits of the course shall not account for the overall CGPA.

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY (CHARUSAT)

Faculty of Technology & Engineering

Teaching & Examination Scheme for B. Tech Programme in CE/CSE/IT Year 2026-27

Course Code	Course Title	Teaching Scheme					Examination Scheme						
		CREDIT				TOTAL HOURS	TH		PR		PRJ		TOTAL
		TH	PR	PRJ	TOTAL		Internal	External	Internal	External	Internal	External	
CEUC301	FUNDAMENTALS OF OPERATING SYSTEM DESIGN	3.00	1.00		4.00	5.00	18/50	18/50	9/25	9/25	-	-	150
CSUC301	MACHINE LEARNING	3.00	1.00		4.00	5.00	18/50	18/50	9/25	9/25	-	-	150
ITUE301	ADVANCED WEB DEVELOPMENT FRAMEWORKS		2.00		2.00	2.00	-	-	9/25	9/25	-	-	50
CEUC302	THROTY OF COMPUTATION	3.00			3.00	3.00	18/50	18/50	-	-	-	-	100
CEUI301	SUMMER INTERNSHIP		3.00		3.00	6.00	-	-	27/75	27/75	-	-	150
CEUP301	PROJECT - I		2.00		2.00	2.00	-	-	9/25	9/25	-	-	50
HSUA301	COMMUNICATION AND SOFT SKILLS		2.00		2.00	2.00	-	-	9/25	9/25	-	-	50
					20.00	25.00							700

ELECTIVE	Course Code	Course Title	Teaching Scheme					Examination Scheme						
			CREDIT				TOTAL HOURS	TH		PR		PRJ		TOTAL
			TH	PR	PRJ	TOTAL		Internal	External	Internal	External	Internal	External	
	CSUA301	INTRODUCTION TO RESEARCH METHODOLOGY		2.00		2.00	2.00	-	-	9/25	9/25	-	-	50
	CEUA301	COMPETITIVE PROGRAMMING ESSENTIALS		2.00		2.00	2.00	-	-	9/25	9/25	-	-	50
	ITUA301	ENTREPRENEURSHIP SKILLS		2.00		2.00	2.00	-	-	9/25	9/25	-	-	50
	CSUE301	BIG DATA ANALYTICS	2.00	1.00		3.00	4.00	9/25	9/25	9/25	9/25	-	-	100
	CSUE302	INTERNET OF THINGS	2.00	1.00		3.00	4.00	9/25	9/25	9/25	9/25	-	-	100
	CEUE301	DIGITAL IMAGE PROCESSING	2.00	1.00		3.00	4.00	9/25	9/25	9/25	9/25	-	-	100
	CEUE302	FUNDAMENTALS OF GAME DEVELOPMENT	2.00	1.00		3.00	4.00	9/25	9/25	9/25	9/25	-	-	100
	ITUE302	ETHICAL HACKING ESSENTIALS	2.00	1.00		3.00	4.00	9/25	9/25	9/25	9/25	-	-	100
	ITUE303	NETWORK DEFENCE ESSENTIALS	2.00	1.00		3.00	4.00	9/25	9/25	9/25	9/25	-	-	100

B. Tech. (CE/CSE/IT) Programme

SYLLABI (Semester – 5)

Charotar University of Science and Technology (CHARUSAT)
Faculty of Technology and Engineering (FTE)

Subject: Fundamentals of Operating System Design(CEUC301)

Semester: 5th

Teaching Scheme:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/Week	3	1	-	5	4
Marks	100	50	-	150	

Course Pre-requisites:

- Fundamental of Data Structures and Algorithm Analysis
- Digital Electronics
- Computer Organization and Architecture

Course Outcomes:

After successful completion of this course, students will be able to

CO1: Analyse and evaluate operating system design goals (performance, scalability, reliability, security, portability) and justify architectural choices (monolithic, microkernel, hybrid).

CO2: Analyse and compare process, thread, and scheduling algorithms using quantitative performance metrics (throughput, turnaround time, waiting time, response time, fairness), and assess their suitability under diverse workload characteristics.

CO3: Design and evaluate synchronization and concurrency control mechanisms using semaphores, monitors, and lock-free techniques.

CO4: Analyse and evaluate deadlock scenarios using formal models, and design appropriate prevention, avoidance, detection, and recovery strategies considering system performance, resource utilization, starvation, and live-lock constraints.

CO5: Assess and compare memory management and virtual memory designs (paging, segmentation, hybrid schemes, kernel allocators) by analyzing fragmentation, access latency, security, and scalability trade-offs.

CO6: Evaluate and optimize file system and I/O subsystem designs by comparing disk scheduling, buffering, journaling, and log-structured approaches, and justify design decisions based on performance, reliability, and workload.

CO7: Synthesize and critique operating system design principles across specialized domains (Real-time, Embedded, and mobile OS) and propose system-level design improvements.

Bridge Topics/Self-Study/Revisit:

- ✓ Instruction Cycle and Execution, Timer Interrupts, and Context Switching at Hardware level, William Stallings, *Computer Organization and Architecture*, 10th Ed. (pp. 168–210)
- ✓ Operating system functionalities and objectives, and its type
- ✓ Memory hierarchy and address translation, William Stallings, *Computer Organization and Architecture* (pp. 290–320)

- ✓ Memory Layout (i.e. Text, Data, Heap, and Stack) preferably using C language, Kernighan & Ritchie, *The C Programming Language*, 2nd Ed. (pp. 116–143)
- ✓ Locking (DBMS), Silberschatz, Korth & Sudarshan, *Database System Concepts*, 7th Ed
- ✓ Cache vs virtual memory, William Stallings, *Computer Organization and Architecture* (pp. 320–350)
- ✓ Hardware synchronization primitives, William Stallings, *Computer Organization and Architecture* (pp. 338–370)
- ✓ File system and its Indexing structure, Silberschatz, Korth & Sudarshan, *Database System Concepts* (pp. 660–690)

Syllabus:

Unit No.	Topic	Hrs.
1	Operating System Design, Processes and Threads	4
	- Design goals: Performance, Scalability, Reliability, Security, Portability and Interoperability - Kernel architectures: Monolithic, Microkernel, Hybrid (design trade-offs) - Process Control Block (PCB) and Thread Control Block (TCB) internals - Process and Thread switching its overhead and its performance evaluation - Kernel vs. user-level threads: design and performance implications	
2	CPU Scheduling and Performance Evaluation	4
	- Scheduling metrics and workload characterization - Overview of CPU scheduling algorithms: FCFS, SJF, RR, Priority - Multilevel and real-time scheduling (EDF, RMS) - Multi-level queue and feedback scheduling	
3	Synchronization and Concurrency Control Design	7
	- Race Conditions, Critical section problem - Semaphores, mutexes, monitors, condition variables - Memory consistency and visibility models - Lock-based vs. lock-free designs	
4	Deadlock Modeling and System-level Handling	4
	- Deadlock modelling. Deadlock Prevention and Avoidance strategies - Detection and Recovery trade-offs - Deadlock Vs Starvation – Resources trade-off - Starvation and livelock	
5	Memory Management System Design	5
	- Overview of Memory fragmentation. - Paging, Segmentation, Paging with Segmentation. - Kernel memory allocators (buddy, slab) - Paging vs. Segmentation – Performance Evaluation - Security under paging and segmentation	
6	Virtual Memory and Page Replacement Design	4
	- TLB design and performance - Overview of Page replacement algorithms (FIFO, LRU, Optimal) - Thrashing and working set model	
7	File System Design and Performance	5

	• File system abstractions and implementation • Journaling vs. log-structured file systems • Free space management • Performance evaluation of file system designs. • Security aspect of the file system.	
8	I/O Subsystem and Storage System Design • Overview of Interrupt handling and DMA design • Buffer cache and I/O scheduling • Overview of Disk Scheduling algorithms (FCFS, SSTF, SCAN, C-SCAN, LOOK) • Disk scheduling under mixed workloads • Storage/Retrieval performance trade-offs	4
9	Specialized Operating Systems: Design and Performance Overview • RTOS: Kernel design, scheduling, priority inversion, performance metrics • Embedded OS: Introduction, kernel architecture, task management and scheduling, memory allocation, inter-task communication, performance metrics • Mobile OS: Constraints and design goals, power aware scheduling, memory management, security and sandboxing models, performance metrics	4

Recommended Study Material:

❖ Text books:

1. Operating Systems: Internals and Design Principles by William Stallings, 9th Edition, Pearson, 2021
2. Operating System Concepts by Silberschatz, Galvin, Gagne, 10th Edition, Wiley, 2025
3. Modern Operating Systems by Tanenbaum and Bos, 5th Edition, Pearson, 2022

❖ Reference Books:

1. Operating Systems: Design and Implementation by Tanenbaum, Woodhull, 3rd Edition, Prentice Hall, 2006.
2. Linux Kernel Development by Robert Love, 3rd Edition, Addison-Wesley Professional, 2010.
3. The Design of the UNIX Operating System by Maurice J. Bach, 1st Edition, Prentice Hall, 1986.
4. Real-Time Systems by Jane W. S. Liu, 1st Edition, Prentice Hall, 2000
5. Embedded Android by Karim Yaghmour, 1st Edition, O'Reilly Media, 2013

Instructions to Subject Teachers:

1. Subject Teachers must prepare Lesson Plan in such a way so that internal working level can be address in detail rather than only user-level abstractions/overview so that clear linkage between topics and experimental implementation including performance analysis and skills can be measured. Following sub-points must be taken into consideration.
 - 1.1. Emphasize on design goals and trade-offs (performance, scalability, predictability, security, energy efficiency), and kernel internals (PCB, Queues, etc.) during the lesson planning.
 - 1.2. Integrate performance evaluation metrics and workload analysis during scheduling, memory management, and I/O subsystems.

- 1.3. Use comparative analysis of operating system architectures (general-purpose, real-time, embedded, and mobile) to highlight differing design constraints.
2. Subject teachers are advised to relate theoretical concepts to real system behaviour using Linux as primary operating system and any one as secondary operating system.
3. Selection of programming languages/tools/platform/operating system is under the purview of the subject teachers. However, proper approval process must be followed, which includes HoDs, BoS (Chairman), and Dean FTE in advanced before lesson planning.
4. Subject teachers must ensure to connect each topic explicitly to its practical and laboratory experiments mapping with skills as per the course outcomes. Also, practical includes observation, measurement, and interpretation of system behaviour, not only code execution as a part of pre or post laboratory exercise/questions.
5. Teaching Pedagogy, Continuous evaluation as per NEP-2020 guidelines, and Practical List must be prepared in advanced according to Open Book Examination as well as the Course Outcomes addressing specific skills, and must be included in the Lesson Plan, which should be made available to students before commencement of the semester.
6. Conduct all kernel-level experiments on virtual machines or dedicated lab systems to ensure safety and stability. Avoid modification of core kernel source; restrict students to loadable kernel modules only.
7. Practical list must begin with system call-based experiments before introducing kernel-level programming to ensure gradual learning.
8. Appropriate provision must be made to ensure conduction of viva based on each and every practical definition even though laboratory teachers may only be associated during the laboratory hours focusing on design understanding, analysis, and reasoning, not merely on program output.
9. Installation including configuration of different operating systems must be demonstrated and a comparative study must be carried out during the topic discussion.

Practical List:

Practical 1: Exploring OS Kernel Interfaces via System Calls

Use getpid(), getppid(), uname(), sysinfo(). Correlate outputs with /proc entries

Practical 2: Process Creation and Lifecycle Analysis

Use fork(), exec(), wait(). Trace process states using /proc/[pid]/status

Practical 3: Thread Creation and Context Switch Cost

Create threads using POSIX threads. Measure thread vs process context switch time

Practical 4: Scheduler Behavior Using System Calls

Use sched_setscheduler(), nice(). Measure turnaround, response time

Practical 5: Kernel Module to Inspect Process Control Blocks

Traverse task_struct list. Extract scheduling and memory attributes.

Practical 6: Real-Time Scheduling Experiment

Implement periodic tasks using SCHED_FIFO. Observe deadline misses

Practical 7: System Call Based Synchronization

Use POSIX semaphores and mutexes. Demonstrate race conditions and fixes

Practical 8: Kernel-Level Synchronization

Implement spinlocks and mutexes in kernel module. Compare blocking vs busy waiting

Practical 9: Deadlock Demonstration and Detection

Create circular wait using semaphores. Analyse deadlock conditions

Practical 10: Kernel Memory Allocation Analysis

Use kmalloc(), vmalloc(). Observe slab and buddy allocator behaviour

Practical 11: Virtual Memory & Page Fault Measurement

Allocate large memory using mmap(). Observe page faults via /proc/vmstat

Practical 12: File System Behaviour via System Calls

Analyse open(), read(), write(), fsync(). Compare buffered vs direct I/O

Practical 13: Kernel Module for I/O Monitoring

Monitor block I/O requests. Study request merging and scheduling

Practical 14: System-call driven Kernel monitor

Develop a dashboard application to monitor the CPU and memory usage by processes.

Charotar University of Science and Technology (CHARUSAT)
Faculty of Technology and Engineering (FTE)

Subject: Machine Learning (CSUC301)

Semester: 5th

Teaching Scheme:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/Week	3	1	-	5	4
Marks	100	50	-	150	

Course Pre-requisites:

- Programming Engineering Mathematics, specifically Linear Algebra and Probability Theory

Course Outcomes:

By the end of the course, students will be able to:

CO1: Analyze machine learning problems using mathematical concepts to understand model behaviour and learning dynamics.

CO2: Design, Implement and analyse (both inference and interpretation) application specific supervised/unsupervised learning and NN models by selecting appropriate algorithms, hyperparameters, and evaluation metrics.

CO3: solve complex pattern recognition tasks.

CO4: Evaluate and implement reinforcement learning approaches for sequential decision problems by modelling environments and optimizing policies.

CO5: Assess machine learning models with respect to fairness, interpretability, and data privacy considerations to ensure responsible deployment.

➤ **Pre-requisites online video/ courses:**

- Linear Algebra: Basic of Vectors including Projection, Gradient Vector and Gradient Descent, Distance Measures, (Euclidean, Cosine similarity, etc.), Matrix and operations, Tensor, Characteristics Equation and its meaning, Partial Derivatives,
- Probability Theory: Variables (Discrete and Continuous), Conditional Probability, Probability Mass and Density Function, Measures of Standard (Variance, Covariance, and Correlation), Binomial and Normal distribution, Bayes' Theorem, Likelihood, Maximum Likelihood Estimation, Logistic regression
- Optimization: Objective function, Local and Global Minima, Function Derivatives
- Data Preprocessing, Training Vs Test Data

➤ **Self-Study/Further Study components and materials:**

Syllabus:

Unit No.	Topic	Hrs.
1	Mathematical Foundation	6
	• System of Linear Equations • Convex and Non-convex functions • Loss functions and its minimization • Gradient and Gradient Descent (Batch and SGD) • Entropy and Information Gain • Bias and Variance Trade-offs • Cost function trade-offs and Regularization • Activation Functions: Which, When and Why	
2	Supervised Learning	8
	• Model Architecture, Selection, and Hyperparameter (Grid Search, Random Search, Bayesian optimization, etc. and Fine Tuning) • Linear Regression and Regularization • Classification: Logistic Regression, k-NN, SVM, Decision Trees (Entropy, Gini), Random Forests, Ensemble methods, Gradient Boosting, Bayesian methods • Evaluation Metrics and their interpretation/inference: Accuracy, Precision, Recall, F1, Cohen's Kappa, ROC Curve, Log Loss, Correlation Coefficient, MSE, RMSE, R^2, MAPE • Semi-supervised and Self-Supervised and their importance – Pseudo-labeling, Contrastive learning, overview of autoencoders • Responsible AI in Supervised Learning: Fairness metrics, Interpretability, Data privacy and leakage, Federated Learning	
3	Unsupervised Learning	7
	• Clustering: Overview of K-Means, Hierarchical, DBSCAN, Gaussian Mixture Models, Hierarchical clustering • Clustering Evaluation / Validation: Silhouette Score, DBI, Calinski–Harabasz Index, ARI, Elbow • Dimensionality Reduction: PCA including derivation, t-SNE, Reconstruction Error, • Evaluation Metrics and their interpretation/inference: Silhouette Score, Davies–Bouldin Index, Calinski–Harabasz Index, ARI, NMI, Homogeneity Score, Completeness Score, V-measure • Hyperparameter tuning	
4	Neural Networks	6
	• Introduction and Basic Terminologies: Perceptron, Propagation, Linear Classifier, Layers, Loss Functions including Common Losses	

	• Architecture – Convolution Layer, Pooling, • Activation Functions and their trade-off • Backpropagation: Chain Rule, Gradient Computation for 2-layer NN, Trainable parameter calculation • Training Optimization: SGD, Momentum, Adam • Regularization: Dropout, Batch Normalization, Early Stopping • Hyperparameter tuning: Optimization, Initialization, Architecture complexity • Evaluation Metrics and their interpretation/inference: Apart from common others are Learning curves, Convergence rate, Calibration error, Fairness metrics, and application domain specific	
5	Deep Learning Foundation • Introduction: Deep Learning Vs Machine Learning, • Convolution NN: Operations, Pooling • Transfer Learning: Pretrained models (TensorFlow, Keras, VGG, etc.), Fine tuning strategy • Sequence Models: RNN intuition, Vanishing gradient problem, Overview of LSTM/GRU • Evaluation Metrics and their interpretation/inference: Apart from common others are AUC-ROC, Top-K Accuracy, Training Loss, Validation Loss, Convergence Rate, Learning Curves, and domain specific	6
6	Reinforcement Learning Fundamentals • Introduction: Agent, Environment, Reward • Markov Decision Process • Q-Learning - Bellman equation intuition, Q-update rule derivation, Exploration vs Exploitation • Evaluation Metrics and their interpretation/inference: Cumulative Reward, Average Episode Reward, Policy and Value Loss, Episode Length, Regret, Success Rate	5

Recommended Study Material:

❖ Text books:

1. Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow by *Aurélien Géron*, 3rd Edition, O'Reilly

❖ Reference Books:

1. "Pattern Recognition and Machine Learning by Bishop, Christopher M., 1st Edition. Springer, 2006.
2. Machine Learning by Mitchell, Tom M., 1st Edition. McGraw-Hill, 1997.
3. The Elements of Statistical Learning: Data Mining, Inference, and Prediction by Hastie, Trevor, Robert Tibshirani, and Jerome Friedman. 2nd Edition, Springer, 2009.
4. Machine Learning: A Probabilistic Perspective by Murphy, Kevin P., 1st Edition. Cambridge, MIT Press, 2012.

5. Deep Learning by Goodfellow, Ian, Yoshua Bengio, and Aaron Courville, 1st Edition. Cambridge, MIT Press, 2016

Instructions to Subject Teachers:

1. Subject Teachers must prepare Lesson Plan in such a way so that students are able to do manual mathematical derivation of equations (e.g. cost for Gradient) whenever required to develop analytical understanding and inference drawing instead of tool usage.
 - 1.1. Attention must be given by the subject teacher towards comparison of models at least based on performance and fairness using appropriate Evaluation metrics including interpretation and inference from the result.
 - 1.2. Students must be able to understand mathematical trade-off of the Hyperparameters and their impact on inference. Hyperparameter tuning must be demonstrated in the context of quality evaluation of the model including optimization objectives/prediction and model complexity, and pipeline development
 - 1.3. Usage of proper dataset including preprocessing steps must be demonstrated so that comparison among model can be demonstrated.
2. Selection of programming languages/tools/platform/Dataset is under the purview of the subject teachers. However, proper approval process must be followed, which includes HoDs, BoS (Chairman), and Dean FTE in advanced before lesson planning.
3. Subject teachers must ensure to connect each topic explicitly to its laboratory experiments mapping with skills as per the course outcomes. Also, practical includes observation, measurement, and interpretation of model behaviour.
4. Teaching Pedagogy, Continuous evaluation as per NEP-2020 guidelines, and Practical List must be prepared in advanced according to Open Book Examination as well as the Course Outcomes addressing specific skills, and must be included in the Lesson Plan, which should be made available to students before commencement of the semester.
5. Practical Problem definition from scratch should be prioritized before introducing library-based solutions. Each unit should include exercises where students build and train models manually. Further, Practical Problem Definition must have pre and post laboratory exercise/questions addressing to specific higher-order skills. Remembering nature questions are strictly prohibited.
6. Practical Problem Definition should ensure the complete ML pipeline — data preprocessing, model selection, hyperparameter tuning, evaluation, and interpretation – as far as possible.
7. Practical Problem Definition should follow a flow as problem definition → dataset understanding → algorithm derivation → scratch implementation → metric evaluation → visualization/interpretation → Comparison with the library-based implementation → Pre-laboratory questions and post-laboratory questions.
8. Practical Problem Definitions may be either in the extension format so that at the end it satisfies the objectives of mini project or interpedently with non-repetition of the objectives.
9. Appropriate provision must be made to ensure conduction of viva based on each and every practical definition even though laboratory teachers may only be associated during the laboratory hours.
10. Selection of Reference books and Online resources like COURSERA, NPTEL, etc. is under purview of the subject teachers.

Charotar University of Science and Technology (CHARUSAT)
Faculty of Technology and Engineering (FTE)

Subject: Advanced Web Development Framework (ITUE301)

Semester: 5th

Teaching Scheme:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/Week	-	2	-	2	2
Marks	-	50	-	50	

Course Pre-requisites:

- Database concepts, Query optimization, Schema Design, Indexing
- Scripting language Features.

Course Outcomes:

By the end of the course, students will be able to:

CO1: Design and Implement component-based frontend systems using reactive state management and performance optimization strategies for scalable web applications.

CO2: Design and Implement secure RESTful backend services incorporating middleware pipelines, validation layers, and caching mechanisms.

CO3: Select and evaluate appropriate data modeling, migration, encryption, and storage strategies based on workload characteristics and consistency requirements.

CO4: Design scalable web system architectures by applying stateless service design, load balancing, asynchronous processing, and performance profiling techniques.

CO5: Implement cloud-native deployment pipelines using containerization, CI/CD automation, and deployment strategies while monitoring reliability and resource optimization.

CO6: Apply security engineering principles and responsibly integrate AI-based services into web systems while analyzing ethical, privacy, and governance implications.

➤ **Pre-requisites online video / course:**

- Client-Server Architecture
- UI/UX Principles
- State Management
- Schema design principles
- ACID Vs BASE properties

Syllabus:

Unit No.	Topic	Hrs.
1	Frontend and Component Architecture	5
	• Overview of Frontend architecture: MPA, SPA, SSR, SSG • Design Methodology: Responsive, Atomic • Overview: UI Component Libraries • Introduction: Component Architecture • State Management: Reactive Patterns • Routing Concepts • API Integration and Consumption strategies	
2	Frontend Engineering	4
	• Rendering Lifecycle • Lazy Code Loading • WCAG Principles • Overview: PWA • Performance Optimization • Frontend Security (XSS, CSRF, Token Handling) and Automated Testing (Application level and Security level)	
3	Backend Architecture and Data Engineering	9
	• Maturity Model: Richardson • Middleware and Request Lifecycle • RESTful API Design • Data Validation Layers and Error Handling • Schema Enforcement • In-memory Caching Strategies • Secure Storage Principles • Database Migration Principles • Overview: Data Encryption	
4	Scalable Web Systems	11
	• Architecture: Monolithic Vs Microservices • Connection Pooling and Query Caching • Scaling: Horizontal Vs Vertical • Stateless Services	

	• Load Balancing and Fault Tolerance • Response Time and Resource Utilization: Minimization • CDN Fundamentals • Asynchronous Processing and Message Queues - Background Data Processing, Notification • Profile: CPU and Memory Usage, Energy-efficient web systems, Data Transfer Optimization • Performance Load Testing – RPS, Latency, Error Rate • Site Reliability Engineering	
5	Cloud Deployment	7
	• Overview: Cloud Computing, Cloud Native Systems • Introduction: Cloud Service Models – Serverless, IaaS, PaaS • Containerization and OCI Standards • Docker Fundamentals and Docker File Optimization • CI/CD Pipeline: Automated Test Execution, Performance Degradation • Overview: Resource Limits Configuration • Deployment Strategies: Blue-Green, Rolling Updates	
6	Security and AI Integration	4
	• Introduction: Threat Model • Advanced Authentication and Authorization • Overview: HTTPS and TLS • Secure Coding: Introduction and Practices • AI Service Integration and Prompt Engineering (Scalability) • AI-assisted Customized Designing • Introduction: Responsible AI	

Recommended Study Material:

❖ **Text books:**

1. *Web Technologies* by Uttam K. Roy, Oxford University Press
2. *Internet and Web Technologies* by Raj Kamal, McGraw Hill Education
3. *Cloud Computing* by Anthony T. V., et. al, McGraw Hill Education

❖ **Reference Books:**

1. Designing Data-Intensive Applications by Martin Kleppmann
2. Clean Code by Robert C. Martin
3. Distributed Systems: Concepts and Design by George C., et.al., Pearson Education

Instructions to Subject Teachers:

1. Selection of tools/databases/scripting languages/platform is under the purview of the subject teachers. However, Installation and configuration of tools to be used must be demonstrated.
2. Teaching Pedagogy, Continuous evaluation as per NEP-2020 guidelines, and Practical List must be prepared in advanced according to Open Book Examination as well as the Course Outcomes addressing specific skills, and must be included in the Lesson Plan, which should be made available to students before commencement of the semester.
3. Practical Problem Definition must have pre and post laboratory exercise/questions addressing to specific higher-order skills must be clearly mentioned apart from resources to be utilized for the solution. Remembering nature questions are strictly prohibited.
4. Students must demonstrate performance benchmarking for related topics.
5. Appropriate tool may be demonstrated to prepare web related project documentation.
6. At least two expert sessions each of four hours' duration must be conducted addressing advanced features of selected and/or industry demand tools. Prior approval process must be carried out.
7. Selection of Reference books and Online resources like COURSERA, NPTEL, etc. is under purview of the subject teachers.
8. Subject teachers should announce at least one COURSERA/NPTEL certification course inline with syllabus as an advanced knowledge (not exactly similar to syllabus).

Charotar University of Science and Technology (CHARUSAT)
Faculty of Technology and Engineering (FTE)

Subject: Theory of Computation (CEUC302)

Semester: 5th

Teaching Scheme:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/Week	3	-	-	3	3
Marks	100	-	-	100	

Course Pre-requisites:

- Fundamental of Data Structures

Course Outcomes:

By the end of the course, students will be able to:

CO1: Design regular grammar, regular expression, finite automata and apply them to practical text processing tasks such as lexical analysis and pattern matching.

CO2: Design context free grammar and push-down automata and apply them to model nested language structure and parsing in programming and natural languages.

CO3: Design Turing Machine for the formal language problems and demonstrate their role in general purpose text transformation tasks.

CO4: Model and analyze formal languages using automata and grammars, and relate them to real-world applications such as compilers, interpreters and natural language processing.

CO5: Identify whether a given computational / text processing problem is decidable or undecidable and classify it in terms of complexity classes.

CO6: Apply automata theory concepts to the design of system software components such as lexical analyzers, parsers, and text editors.

CO7: Evaluate computational problems arising in text and language processing and justify solutions using formal models and complexity theory.

➤ **Pre-requisites online video/ courses:**

- Set and Subset, Finite and Infinite Set, Set Operations, Power Set, Set Properties.
- Function and Relation- Transitive, Reflexive, Symmetric, Transitive, Equivalence, Recurrence
- Mathematical Induction (Weak, Strong, Structural), and Proof (Direct and Proof by Contradiction and by Contrapositive) Theory
- Permutation and Combination
- Partition and Truth Tables
- Logic – Propositional and Predicate, Connectives, Inference, Quantifiers
- Basic Probability

- Linked List, Tree (Binary, Rooted) and Graph (Direct, Path, Cycle), and Operations on them
- Recursion
- Regular Expression: Mastering Regular Expressions by Jeffrey Friedl

Syllabus:

Unit No.	Topic	Hrs.
1	Language Foundation	4
	• Fundamental Terms: Language, Terminal and Non-terminal Symbol, Token, Finite and Infinite Language, Grammar, Ambiguity, Kleene Star, Pigeonhole Principle, Adjacency matrices, • Production rule • Classification of Language: Regular, Context Free, Context Sensitive, Recursively Enumerable, Decidable, and Free languages • Overview of Closure Properties and Pumping Lemma	
2	Finite Automata	11
	• Introduction – Automata Theory • Deterministic Finite Automata including Extended Transition Function and its minimization • Non-Deterministic Finite Automata including Equivalence of DFA and NFA • Regular Languages: Equivalence and Kleene's Theorem • Conversions and Closure properties • Pumping Lemma – Proof technique and Non-regular language Identification • Lexical Analysis – Token Recognition and Classification, Control Structure Recognition, Class & Object Identifier Recognition, Access Modifier Recognition, Token Stream Processing, Template Parsing, Interpreter Tokenization, Dynamic Syntax Checking • DFA-based Scanners, and UML and DFA equivalence and vice-versa	
3	Context Free Language (CFG)	13
	• Context Free Language and Grammar • Language Generation: Derivation – Leftmost and Rightmost, Sentential Forms, • Parsing • Ambiguity: Multiple Parse trees, Arithmetic Expression, Dangling Problem, Inherent Ambiguity, Precedence Rules, Grammar Restructuring, Elimination Procedure, Removal Technique • CNF: Properties and Conversion • GNF • Pumping Lemma • Closure and Non-Closure • Problems: Membership, Finiteness, and Emptiness	

	• Parsing: Arithmetic Expression, Natural language	
4	Pushdown Automata (PDA) • Introduction • Deterministic Vs Non-deterministic PDA • Equivalence: CFG and PDA • Design of PDA: Balanced Parenthesis, $a^n b^n$ type language • Decidable Vs Recognizable languages, Undecidable Problems • Overview – Protocol Verification, Linear Bounded Automata, Rice's Theorem, Halting Problem	7
5	Turning Machine (TM) • Introduction – TM, Transition function • Variants – Deterministic TM, Non-deterministic TM, Multi-tape TM, and Multi-track TM • Design Problems at Binary addition and String operations	5

Recommended coding Languages and tools:

1. Hypertext Markup Language 5 (HML5)
2. Cascading Style Sheets 3(CSS3)
3. MySQL and MongoDB

Recommended Study Material:

❖ **Text books:**

1. Introduction to the Theory of Computation by Michael Sipser, 3rd Edition, Cengage.
2. Introduction to Language and Theory of Computation, John C. Martin, TMH

❖ **Reference books:**

1. Introduction to the Theory of Computation by Michael Sipser, 3rd edition, Cengage Learning
2. Introduction to computer theory by Deniel I. Cohen, John Wiley & Sons Inc
3. Compilers: Principles, Techniques, and Tools by Aho, Lam, Sethi and Ullman, Dragon
4. Introduction to Automata Theory, Languages, and Computation by Hopcroft, Motwani and Ullman
5. An introduction to Formal Languages and Automata by Peter Linz, 6th edition, Jones & Bartlett Learning

Instructions to Subject Teachers:

1. Subject Teachers must prepare Lesson Plan in such a way so that the mapping of topics among this subject with Text processing and Compiler Construction can be covered as an example in a parallel way.
2. Subject teachers must assign implementation (e.g. Lexical analyzer, Code Analyzer, Template parsing, Fitness problem, Regex engine, Protocol verification, etc.) from scratch of appropriate topics in a group as a part of Continuous Evaluation including comparison, based on appropriate parameters (e.g. Code Correctness and Clarity, Time Complexity, Scalability, Boundary Value conditions, etc.), with solution generated by AI-based tool. Rubric based evaluation is recommended.
3. Selection of tools /platform is under the purview of the subject teachers.
4. Teaching Pedagogy, Continuous evaluation as per NEP-2020 guidelines, and Practical List must be prepared in advanced according to Open Book Examination as well as the Course Outcomes addressing specific skills, and must be included in the Lesson Plan, which should be made available to students before commencement of the semester.
5. Practical Problem Definition must have pre and post laboratory exercise/questions addressing to specific higher-order skills must be clearly mentioned apart from resources to be utilized for the solution. Remembering nature questions are strictly prohibited.
6. At least two expert sessions each of four hours' duration must be conducted addressing current industry trend. Prior approval process must be carried out.
7. Selection of Reference books and Online resources like COURSERA, NPTEL, etc. is under purview of the subject teachers.
8. Subject teachers should announce at least one COURSERA/NPTEL certification course in line with syllabus as an advanced knowledge (not exactly similar to syllabus) focused on specific topics related to *Basic of Text Processing* in the beginning of the semester and ensure to provide necessary guidance to those students whose carrier path is associated with this subject.

Charotar University of Science and Technology (CHARUSAT)
Faculty of Technology and Engineering (FTE)

Subject: Summer Internship (CEUI301)

Semester: 5th

Teaching Scheme:

Teaching Scheme	Project	Practical	Tutorial	Total	Credit
Hours/Week	90	-	-	90	6
Marks	150	-	-	150	

Course Outcomes:

By the end of the course, students will be able to:

1. **CO1:** Ability to integrate existing and new technical knowledge for industrial application.
2. **CO2:** Executing work with team and teammates from other disciplines
3. **CO3:** Get practices and experience related to professional and ethical issues in the work environment
4. **CO4:** Experience of demonstrating the impact of the internship on their learning and professional development.
5. **CO5:** Understanding of lifelong learning processes through critical reflection of internship experiences.

Syllabus:

Unit No.	Topic
1	Instructional Method and Pedagogy
	• Summer internship shall be at least 90 hours during the summer vacation only. • Department/Institute will help students to find an appropriate company/industry/organization for the summer internship. • The student must fill up and get approved a Summer Internship Acceptance form by the company and provide it to the Coordinator of the department within the specified deadline. • Students shall commence the internship after the approval of the department Coordinator. Summer internships in research centers is also allowed. • During the entire period of internship, the student shall obey the rules and regulations of the company/industry/organization and those of the University.

	Due to inevitable reasons, if the student will not able to attend the internship for few days with the permission of the supervisor, the department Coordinator should be informed via e-mail and these days should be compensated later.
2	Format of Summer Internship Report - The report shall comply with the summer internship program principles. Main headings are to be centered and written in capital boldface letters. Sub-titles shall be written in small letters and boldface. The typeface shall be Times New Roman font with 12pt. All the margins shall be 2.5cm. The report shall be submitted in printed form and filed. An electronic copy of the report shall be recorded in a CD and enclosed in the report. Each report shall be bound in a simple wire vinyl file and contain the following sections: - Cover Page - Page of Approval and Grading - Abstract page: An abstract gives the essence of the report (usually less than one page). Abstract is written after the report is completed. It must contain the purpose and scope of internship, the actual work done in the plant, and conclusions arrived at. - TABLE OF CONTENTS (with the corresponding page numbers) LIST OF FIGURES AND TABLES (with the corresponding page numbers) - DESCRIPTION OF THE COMPANY/INDUSTRY/ORGANISATION: Summarize the work type, administrative structure, number of employees (how many engineers, under which division, etc.), etc. Provide information regarding - Location and spread of the company - Number of employees, engineers, technicians, administrators in the company - Divisions of the company - Your group and division - Administrative tree (if available) - Main functions of the company - Customer profile and market share - INTRODUCTION: In this section, give the purpose of the summer internship, reasons for choosing the location and company, and general information regarding the nature of work you carried out. - PROBLEM STATEMENT: What is the problem you are solving, and what are the reasons and causes of this problem. - SOLUTION: In this section, describe what you did and what you observed during the summer internship. It is very important that majority of what you write should be based on what you did and observed that truly belongs to the company/industry/organization.

	• CONCLUSIONS: In the last section, summarize the summer internship activities. Present your observations, contributions and intellectual benefits. If this is your second summer internship, compare the first and second summer internships and your preferences. • REFERENCES: List any source you have used in the document including books, articles and web sites in a consistent format. • APPENDICES: If you have supplementary material (not appropriate for the main body of the report), you can place them here. These could be schematics, algorithms, drawings, etc. If the document is a datasheet and it can be easily accessed from the internet, then you can refer to it with the appropriate internet link and document number. In this manner you don't have to print it
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Recommended Study Material:

❖ Text books:

1. "Java: The Complete Reference, Thirteenth Edition", by Herbert Schildt, McGraw Hill.
2. "OCP Oracle Certified Professional Java SE 17 Developer (Exam 1Z0-829) Programmer's Guide", by Khalid A. Mughal, Vasily A. Strelnikov, Oracle Press.

❖ Reference Books:

1. Books, Magazines & Journals of related topics

❖ Web Material:

1. www.ieeeexplore.ieee.org
2. www.sciencedirect.com
3. www.elsevier.com
4. <http://spie.org/x576.xml>

❖ Software:

5. ASP.NET
6. PYTHON/MATLAB
7. PHP
8. ANDROID/IOS
9. FLUTTER
10. NODE/REACT NATIVE

Charotar University of Science and Technology (CHARUSAT)
Faculty of Technology and Engineering (FTE)

Subject: Introduction to Research Methodology (CSUA301) (University Elective - I)

Semester: 5th

Teaching Scheme:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/Week	-	2	-	2	2
Marks	-	50	-	50	

Course Pre-requisites:

- Basic Mathematics and Statistics, Subject-Specific Knowledge, Computer Literacy.

Course Outcomes:

By the end of the course, students will be able to:

CO1: Explain the fundamental concepts and principles of research methodology, including the scientific method and various research paradigms.

CO2: Design a comprehensive research plan, including selecting appropriate research strategies, methodologies, and data collection techniques.

CO3: differentiate between and apply quantitative and qualitative research methods appropriately based on the research question.

Syllabus:

Unit No.	Topic	Hrs.
1	Introduction to Research	6
	Meaning, Objective and Motivation in Research. Types of Research: Descriptive vs. Analytical, Applied vs. Fundamental, Quantitative vs. Qualitative, Conceptual vs. Empirical, Concept of Applied and Basic Research Process, Criteria of Good Research.	
2	Literature Review	10
	Objectives of Review of Literature, Importance of Literature Review in Defining a Problem, Steps in Conducting Literature Review - Guidelines on Style, Mechanics and language usage - Writing up to the literature Reviewed - Some Examples	
3	Technical Writing	8
	Introduction, Steps of Research, Types of Research Article, Structure of Review Research Article, Review Article,	

	• Publication Ethics, Length of Review Article	
4	Tools for Research • Methods to search required information effectively, Reference • Management Software like Zotero /Mendeley, Software for paper formatting like LaTeX / MS Office, Overleaf.	6

Recommended Study Material:

❖ **Text books:**

1. C.R. Kothari and Gaurav Garg, “Research Methodology: Methods and Techniques”, New Age International (P) Ltd., Publishers, Fourth Multi Colour Edition, 2020.

❖ **Reference books:**

1. Research Methodology And Statistical Tools - P. Narayana Reddy And G.V,R.K, Acharyulu, 1st Edition, Excel Books, New Delhi, 2008.

Charotar University of Science and Technology (CHARUSAT)
Faculty of Technology and Engineering (FTE)

Subject: Competitive Programming Essentials (CEUA301) (University Elective – I)

Semester: 5th

Teaching Scheme:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/Week	-	2	-	2	2
Marks	-	50	-	50	

Course Pre-requisites:

- Data Structures and Algorithms
- Familiarity with a programming language.

Course Outcomes:

By the end of the course, students will be able to:

CO1: apply advanced algorithmic techniques to solve complex problems.

CO2: develop the ability to analyze and optimize the time complexity of algorithms.

CO3: gain mathematical skills crucial for algorithmic problem-solving

CO4: design efficient algorithms and develop strong problem-solving skills.

CO5: Compete in competitive programming contests and coding interviews.

Syllabus:

Unit No.	Topic	Hrs.
1	Introduction to competitive programming and techniques	2
	• Meaning Of Competitive Programming, Programming Techniques (Language Features Input/Output, Numbers, Code Shortening), Recursive Algorithms (Generating Subsets, Generating Permutations, Backtracking), Bit Manipulation (Bit Operations, Set Operations).	
2	Time Efficiency	4
	• Common Time Complexity and Calculation Rules, Examples (Maximum Sub-Array Sum Problem, 2 Queens), Algorithm Analysis, Situational Awareness, Touch Typing Exercise to • Improve Typing Speed.	
3	Mathematics for competitive programming	4

	• Geometry (Points and Lines), Closest Pair Problem, Number Theory, Sum Formulae and Logarithms, Inclusion and Exclusion, Combinatorics	
4	Linear data structures and Libraries	5
	• Static Array, Dynamic Array, Boolean Array, LinkedList, Stack, Queue, Double-Ended Queue, Exercise Problems.	
5	Non-linear data structures and Libraries	5
	• Binary Search Tree (Map/Set), Heap (Priority Queue), Hash Table (Unordered Map), Graph (Adjacency List and Matrix, Edge List), Exercise Problems.	
6	Sorting, searching and algorithm design	6
	• Complete Search (Iterative and Recursive), Linear and Binary Search, Sorting Algorithm and Library, Two Pointers, Nearest Smaller/Greater Element, Sliding Window, Ternary Search, Minimizing Sum, hamming distance, counting sub grids, • reachability in graphs	
7	String processing algorithms	4
	• Basic string processing, string processing problem categories, string matching, longest common subsequence, levenshtein distance	

Recommended Study Material:

❖ **Text books:**

1. competitive programming by Steven & Felix Halim
2. Guide to Competitive Programming: Learning and Improving Algorithms Through Contests.

Charotar University of Science and Technology (CHARUSAT)
Faculty of Technology and Engineering (FTE)

Subject: Big Data Analytics (CSUE301) (University Elective – I)

Semester: 5th

Teaching Scheme:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/Week	2	1	-	4	3
Marks	50	50	-	100	

Course Pre-requisites:

- Foundation of Data Science and Analytics
- Database Management System
- Engineering Mathematics

Course Outcomes:

By the end of the course, students will be able to:

CO1: Analyze performance trade-offs of large-scale data systems and evaluate distributed computing models for big data applications.

CO2: Design and evaluate distributed storage solutions using distributed file systems and NoSQL databases considering scalability, fault tolerance, and query performance.

CO3: Apply data processing techniques to implement scalable data analytics workflows and performance evaluation.

CO4: Design real-time data analytics pipelines for high-velocity data.

CO5: Implement scalable machine learning algorithms and analytical models for large-scale datasets.

CO6: Design and implement interactive visualization systems and analytical dashboards for extracting actionable insights from large-scale datasets.

Syllabus:

Unit No.	Unit/Topic Details	Hours
1	Big Data Fundamentals	4
	• Characteristics and Challenges • Data Life Cycle in Large Scale System • Distributed Computing Models - Shared-nothing architecture, BSP, Stream Processing, Graph Processing, Map-Reduce	

	• Big Data Architecture – Layered Big Data, Stream Processing, Lambda, Kappa and Data Lake • Performance Metrics - Latency, Throughput, Scalability, Resource utilization • Consistency Vs Availability Vs Partition tolerance , and Storage Vs Computation Vs Communication cost trade-off • Responsible AI and Data Governance Principles	
2	Distributed Storage Systems • Overview: Distributed File Systems - Parallelism, Replication, Data partitioning, Fault tolerance, Data Locality and Data Placement strategies • Hadoop Distributed File System – Operations, Block Management and Data Partitioning strategies (Hash and Range) • NoSQL Models: Key-Value Stores, Document Stores, Column and Graph Databases • Query – Indexing strategies, Distributed query execution • Storage Performance Optimization – Log-structured storage, Data Layout optimization, Overview of Compression and Encoding techniques	5
3	Parallel and Distributed Data Processing • Parallelism and its types • Data Partitioning strategies – Horizontal, Vertical, Hash, Data Locality and Partition placement • Data Processing Operations - Parallel Aggregations and Reduction Operations, Filtering and Transformation, Distributed Joins • Distributed Query Optimization – Minimizing Communication Cost, Joint optimization techniques, Broadcast Joins and Shuffle operations • Cluster Resource Management - Job Scheduling (Fair Vs Capacity scheduling), Resource Allocation strategies • Load Balancing – Static Vs Dynamic load balancing, Data Mitigation strategies • Fault Tolerance – Checkpoints, Re-execution and Recovery • Parallelism and its types	5
4	In-Memory Data Processing • Core Data abstractions – RDD, Data Frames, Data Sets • Operations – Transformations, Actions, and Lazy Evaluation • Performance Optimization – Caching and Persistence, Partition Tuning, Shuffle optimization	3
5	Streaming Analytics and Real-Time Data Processing	5

	• Stream processing vs Batch processing Vs Real-Time processing Vs On-line processing • Event streams processing models • Window-based Stream Operations - Tumbling, Sliding, Landmark • Event time vs Processing time Trade-off • Approximation Query Processing - Count-Min Sketch, Bloom Filters, HyperLogLog • Stream Mining Algorithms – Online Classification, Online Clustering, and Change & Novelty Detection	
6	Scalable Machine Learning – Big Data • Challenges • Distributed training – Data parallelism and Model parallelism • Scalable Optimization models - Mini-batch gradient descent, Distributed stochastic gradient descent • Scalable Analytics Algorithms – Clustering (Parallel K-Means, Distributed hierarchical clustering, Density-based clustering), Classification (Distributed decision trees, Large-scale logistic regression), Graph analytics	4
7	Scalable Visualization and Analytics • Scalable visualization Principles - Data aggregation, Sampling and approximation techniques, Multi-resolution visualization • Interactive analytics systems - Dashboard analytics, Visual exploration, Geospatial analytics • Visualization – Temporal analysis, Spatial analysis, Graph visualization • Analytical dashboards – Design Principles, Actionable insights Vs Data dumping, User-driven exploration, Visual Encoding Techniques (Temporal and Spatial pattern analysis, Tree Maps, Sunburst charts, Dendrograms), Narrative visualization, Decision-support visualization	4

Recommended Study Material:

❖ **Text books:**

1. Big Data, Big Analytics by Michael Minelli, Michele Chambers, Ambiga Dhiraj, Wiley Publication
2. Mining of Massive Datasets by Jure Leskovec, Anand Rajaraman, and Jeffrey David Ullman, Publisher: Cambridge University Press

Recommended Study Material:

❖ Text books / Reference Books:

1. Streaming Systems: The What, Where, When, and How of Large-Scale Data Processing" by Tyler Akidau, Slava Chernyak, and Reuven Lax, Publisher: O'Reilly Media
2. Designing Data-Intensive Applications: The Big Ideas Behind Reliable, Scalable, and Maintainable Systems by Martin Kleppmann, Publisher: O'Reilly Media
3. Hadoop: The Definitive Guide by Tom White, O'Reilly Publications.
4. Spark: The Definitive Guide – Big Data Processing Made Simple by Bill Chambers and Matei Zaharia, O'Reilly Publication

Instructions to Subject Teachers:

1. Subject Teachers must prepare Lesson Plan in such a way so that students are able to develop reasoning and analytical understanding and inference drawing by way of different-tradeoff.
2. Appropriate applications (e.g. Log Analytics, Fraud Detection, Recommended systems, etc.) along with analysis must be consider for understanding of different trade-off and must be included in the Lesson plan document.
3. Selection of programming languages/tools/platform/Dataset is under the purview of the subject teachers. However, proper approval process must be followed, which includes HoDs, BoS (Chairman), and Dean FTE in advanced before lesson planning.
4. Subject teachers must ensure to connect each topic explicitly to its laboratory experiments mapping with skills as per the course outcomes.
5. Teaching Pedagogy, Continuous evaluation as per NEP-2020 guidelines, and Practical List must be prepared in advanced according to Open Book Examination as well as the Course Outcomes addressing specific skills, and must be included in the Lesson Plan, which should be made available to students before commencement of the semester.
6. Practical Problem Definition must have pre and post laboratory exercise/questions addressing to specific higher-order skills. Remembering nature questions are strictly prohibited.
7. Practical Problem Definitions may be either in the extension format so that at the end it satisfies the objectives of mini project or interpedently with non-repetition of the objectives.
8. Appropriate provision must be made to ensure conduction of viva based on each and every practical definition even though laboratory teachers may only be associated during the laboratory hours.
9. Selection of Reference books and Online resources like COURSERA, NPTEL, etc. is under purview of the subject teachers.

Charotar University of Science and Technology (CHARUSAT)
Faculty of Technology and Engineering (FTE)

Subject: Internet of Things (CSUE302) (University Elective-I)

Semester: 5th

Teaching Scheme:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/Week	2	1	-	4	3
Marks	50	50	-	100	

Course Pre-requisites:

- Foundation of Data Science and Analytics
- Database Management System
- Engineering Mathematics

Course Outcomes:

By the end of the course, students will be able to:

CO1: Understand the fundamental of IoT architecture, ecosystem, and key components.

CO2: Apply IoT communication protocols such as MQTT, CoAP, and HTTP in real-time applications.

CO3: Evaluate IoT system performance with respect to scalability, security, and reliability.

CO4: Design and develop IoT applications across domains like smart cities, healthcare, agriculture, and industry.

CO5: Integrate IoT devices with cloud platforms for data storage, monitoring, and analytics.

Syllabus:

Unit No.	Unit/Topic Details	Hours
1	IoT Architecture And Components	5
	• IoT ecosystem and architecture • Perception Layer: Sensors and Actuators • Network Layer: Communication protocols • Middleware Layer: Data processing and storage • Application Layer: User interfaces, cloud services	
2	IoT Communication Technologies	5

	• RFID, NFC, Bluetooth, ZigBee • Wi-Fi, LoRa, Sigfox, NB-IoT • Cellular Connectivity (2G/3G/4G/5G) • Fundamentals of Ad-hoc Network	
3	Low Power Iot Modules • Introduction to Low Power IoT Modules • Common platforms: Arduino, Raspberry Pi, ESP32 • Programming basics: C, Python, MicroPython • Understanding of GPIO, PWM, ADC, UART, I2C, SPI	5
4	IoT Protocols & IoT Cloud • MQTT, CoAP, HTTP, WebSockets • IPv6, 6LoWPAN • RESTful APIs for IoT • Introduction to cloud platforms: AWS IoT, Azure IoT, Google Cloud IoT • Data logging and visualization: ThingSpeak, Blynk, Firebase • Difference between Edge Computing and Cloud Computing	5
5	IoT Security & Privacy • Security threats and challenges in IoT • Encryption, authentication, access control • Secure communication protocols • Case studies on IoT breaches	5
6	Data Analytics For IoT • Data collection and preprocessing • Sensor data analytics • Visualization tools (Grafana, Power BI) • Introduction to machine learning for IoT	5

Recommended Study Material:

❖ **Text books:**

1. Internet of Things: A Hands-On Approach by Arshdeep Bahga, Vijay Madisetti –MGHill, Publication
2. Getting Started with the Internet of Things by Cuno Pfister, O'Reilly

❖ **Reference books:**

1. Designing the Internet of Things by Adrian McEwen, Hakim Cassimally, Wiley
2. Internet of Things: Architectures, Protocols and Standards by Simone Cirani, Gianluigi Ferrari, Marco Picone, Luca Veltri, Wiley.

Charotar University of Science and Technology (CHARUSAT)
Faculty of Technology and Engineering (FTE)

Subject: Digital Image Processing (CEUE301) (University Elective -I)

Semester: 5th

Teaching Scheme:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/Week	2	1	-	4	3
Marks	50	50	-	100	

Course Pre-requisites:

- Engineering Mathematics
- Introduction to Computer Graphics

Course Outcomes:

By the end of the course, students will be able to:

CO1:Analyse digital image formation models and sampling processes to evaluate the impact of spatial and intensity resolution on image quality.

CO2:Design and implement spatial domain image enhancement algorithms and interpret their effects using evaluation metrics.

CO3:Apply frequency-domain transformations to perform filtering and analyse the relationship between spatial and spectral image representations.

CO4:Develop restoration models to mitigate degradation caused by noise and system imperfections using appropriate statistical filtering techniques.

CO5:Evaluate image compression techniques by analyzing redundancy removal, compression ratio, and reconstruction quality.

CO6:Design segmentation pipelines using thresholding, edge-based, region-based, and morphological methods and assess their performance using segmentation metrics.

Syllabus:

Unit No.	Topic	Hrs.
1	Introduction to competitive programming and techniques	4
	• Components and Steps • Analog image Vs Digital image, Gray-level images Vs Binary images Vs Grid-based • Overview: Sampling, Quantization, Resolution (Spatial, Intensity), Intensity	

	• DFT and Convolution theorem • Pixels representation – Connectivity, Adjacency, Distance measures • Adjacency Vs Connectivity • Image Operations: Zooming and Shrinking, Interpolation	
2	Time Efficiency • Operations – Arithmetic, Logical, Set • Overview: Filtering and Transformation, Point Processing, Convolution and Correlation • Intensity Transformation – Image negation, Contrast Stretching, • Inverse Log Transformation, Power-law Transformation, Gray and • Bit-plan Slicing Histogram-based Transformation - Equalization, Specification Spatial Domain Enhancement – Filtering Fundamentals, Smoothing filters (Mean, Gaussian, Order-statistic), Sharpening filters (Laplacian, Gradient-based), Edge enhancement Frequency Domain Enhancement – Transform-Domain Representation, Smoothing and Sharpening filters (Low-pass, High-pass, Band-pass), Homomorphic filtering	8
3	Mathematics for competitive programming • Overview: Restoration and Noise models (Gaussian, Impulse, Rayleigh, Exponential, and Periodic) • Spatial Domain Restoration – Mean, Order-statistic, Adaptive • Frequency Domain Restoration – Image Degradation (PSF and Linear), Periodic Noise Removal (Band-reject filters, Band-pass filters, Notch filters, Notch Reject filters), Inverse Filtering, Wiener Filtering, Constrained Least Squares Filtering	5
4	Non-linear data structures and Libraries • Radiance, Luminance, Brightness • Colour Models - RGB, HSI, CMY, CMYK, CIE • Data Compression Vs Image Compression • Data Redundancy Types: Coding, Interpixel, Psychovisual • Compression Types: Lossy Vs Lossless • Techniques: RLE, Huffman coding and encoding, LZW, DCT, DWT, Vector Quantization, Quantization, Wavelet-based	4
5	Sorting, searching and algorithm design • Segmentation Vs Classification Vs Detection • Pixel-based Vs Region-based segmentation • Thresholding Techniques – Global, Adaptive, Otsu's, • Edge-based Segmentation - Gradient-based edge detection, Canny edge detection, LoG, Sobel and Prewitt operators • Region-based Segmentation - Region growing, Splitting and Merging, Homogeneity criteria	5

	• Overview: Clustering-based Segmentation, Watershed segmentation Texture-based Segmentation, Graph-based Segmentation, Morphological Segmentation • Segment Quality Evaluation – IoU, Dice coefficient, Boundary matching metrics, SSIM	
6	String processing algorithms • Structuring Elements - Shape, Size, Orientation • Erosion and Dilation - Duality property • Morphological gradient - Top-hat transform, Bottom-hat transform • Skeletonization - Thinning and thickening • Watershed Segmentation - Marker-controlled, Morphological reconstruction • Morphological reconstruction – Marker Vs Mask, Geodesic Erosion and Dilation	4

Recommended Study Material:

❖ **Text books:**

1. Digital Image Processing by Rafael C Gonzalez, Richard E Woods, Pearson Education
2. Digital Image Processing by W. Pratt, Wiley Publication

❖ **Reference books:**

1. Fundamentals of Digital Image Processing by A.K. Jain, PHI
2. Digital Image Processing and Computer Vision by Milan Sonka, Thomson publication
3. Digital Image Processing by S. Jayaraman, S Esakkirajan and T Veerakumar, McGraw Hill Education (India) Private Limited
4. Digital Image Processing by S. Sridhar, Oxford University Press

Charotar University of Science and Technology (CHARUSAT)
Faculty of Technology and Engineering (FTE)

Subject: Fundamental of Gaming Development (CEUE302) (University Elective – I)

Semester: 5th

Teaching Scheme:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/Week	-	4	-	4	2
Marks	-	50	-	50	

Course Pre-requisites:

- Engineering Mathematics
- Introduction to Computer Graphics
- Programming Fundamentals using C/C++/Java
- Object-Oriented Programming Concepts
- Data Structures and Algorithms
- Basics of Linear Algebra and Vectors
- Fundamentals of Physics (Motion, Force, Velocity)
- Problem Solving and Logical Thinking
- Basics of Multimedia and Audio Concepts
- Introduction to Artificial Intelligence Concepts

Course Outcomes:

By the end of the course, students will be able to:

CO1:Analyze and interpret the architecture of game systems, including component-based structures for real-time interactive applications.

CO2:Apply rendering principles to design and evaluate transformations, lighting models, and texture mapping techniques for game environments.

CO3:Evaluate different motion models, collision detection techniques, and response strategies to ensure realistic object interactions in games.

CO4:Design animation systems and player interaction models

CO5:Evaluate game system performance considering memory management, resource allocation, audio integration, and optimization strategies.

CO6:Design conceptual models for intelligent and adaptive game components using machine learning concepts and agent-based approaches.

CO7:Develop and integrate core game components to build game prototypes demonstration.

Syllabus:

Unit No.	Topic	Hrs.
1	Game Architecture and Foundation	7
	• Important Primitives: Frame Rate, Game Timing, Time Step, Illumination • Overview: Game Architecture including Game engine's core components • Game Development Lifecycle • Game Loop • Game engine architecture layers • Event Driven Systems • Input Devices and Event Handling - Continuous vs Discrete input • Overview: Real-time simulation, Deterministic Vs Non-Deterministic Simulation • Overview: Graphics Pipeline • Game programming patterns - Object pool, Observer, State pattern • Overview: Transformations - 2D and 3D, Vector and Coordinate Systems • Distance and Intersection computations • Spatial Reasoning for Movement and Interactions	
2	Rendering Fundamentals • Player Control Models • Scene representation and Scene graph • Object representation – 2D and 3D • Viewing transformations • Lighting Primitives, and Model – Ambient, Diffuse Reflection, Specular Reflection, Phong Reflection, Local Vs Global Illumination • Texture mapping – UV mapping, Texture Vs Object Space, Mapping techniques, Texture Sampling and Filtering (Nearest-neighbour, Bilinear, Mipmapping), Memory Management for Textures • Frame buffers and Rendering cycles • Animation – Keyframe, Skeletal, Interpolation techniques • Animation Blending and State-based animation control	8
3	Object Motion and Interactions	7

	• Collision primitives - Gravity, Acceleration, Velocity, Contact Points, Collision Vector, Penetration depth • Motion equations and Numerical integration • Collision detection techniques – Bounding Volume, Intersection Tests, Spatial Partitioning, Discrete Vs Continuous Collision detection • Collision response strategies – Basic Response, Impulse-based, Constraint-based • Simple rigid body simulation	
4	Game Audio and Optimization	4
	• Audio Pipeline and Music Integration • Spatial Audio • Event-based audio triggering • Game state management • Resource and Memory management • Optimization strategies - Update loops, Object pooling • Introduction: Multiplayer and Networking concepts	
5	ML and Agentic AI	4
	• Search Space – A* search algorithm, Pathfinding, Decision trees • ML in Game Development – Rule-based Vs Learning-based • Intelligent agents and Environment interaction • Perception Vs Decision Vs Action cycle • Reactive vs Deliberative agents • Goal-driven behaviour and Learning from player behaviour • Reinforcement learning in Games	

Recommended Study Material:

❖ **Text books:**

1. Game Programming Patterns by Robert Nystrom (2014)
2. Fundamentals of Game Development by Heather Maxwell Chandler, Jones & Bartlett Learning (2014)
3. Beginning C++ Through Game Programming, 4th Edition by Michael Dawson (2018)

❖ **Reference books:**

1. Game Engine Architecture, 3rd Edition by Jason Gregory (2018)
2. Artificial Intelligence for Games, 3rd Edition by Ian Millington and John Funge (2016)
3. Real-Time Rendering, 4th Edition by Tomas Akenine-Möller, Eric Haines, and Naty Hoffman (2018)
4. Mathematics for 3D Game Programming and Computer Graphics, 3rd Edition by Eric Lengyel (2011)
5. Unity in Action, 2nd Edition by Joe Hocking (2018)

Charotar University of Science and Technology (CHARUSAT)
Faculty of Technology and Engineering (FTE)

Subject: Ethical Hacking Essentials (ITUE302) (University Elective – I)

Semester: 5th

Teaching Scheme:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/Week	2	2	-	4	3
Marks	50	50	-	100	

Course Pre-requisites:

- Database concepts, Query optimization, Schema Design, Indexing
- Scripting language Features.

Course Outcomes:

By the end of the course, students will be able to:

CO1: Explain key information security challenges, laws, and regulations, and describe ethical hacking concepts and methodologies.

CO2: Identify and analyze security threats, malware types, and vulnerabilities, and perform vulnerability assessments and recommend appropriate countermeasures.

CO3: Analyze security threats across network, web application, wireless, mobile, IoT, OT, and cloud environments, and apply appropriate defense techniques.

CO4: Explain and apply penetration testing fundamentals, phases, methodologies, and guidelines to assess and report security risks effectively.

CO5: Apply ethical hacking tools and techniques in a controlled environment to identify, analyze, and mitigate security vulnerabilities while adhering to legal and ethical guidelines.

➤ **Pre-requisites online video / course:**

- Introduction to Information Security - https://www.cengage.com/resource_uploads/downloads/1111138214_259146.pdf
- Basic Concepts of Networking (Part I) - <https://www.youtube.com/watch?v=-iShap4w4S8>
- Basic Concepts of Networking (Part II) - <https://www.youtube.com/watch?v=jp0JdeeFUG0>
- TCP/IP Protocol Stack (Part I) - <https://www.youtube.com/watch?v=EsJVCgefqnY>
- TCP/IP Protocol Stack (Part II) - <https://www.youtube.com/watch?v=atVnsf6PhO0>

Syllabus:

Unit	Topics	Hours
1.	Information Security Fundamentals	2
	• The need for security • The elements of information security • The security, functionality, and usability triangle • The motives, goals, and objectives of information security attacks • The classification of attacks • The information security attack vectors • Know about the information security laws and regulations	
2.	Ethical Hacking Fundamentals	2
	• The cyber kill chain methodology • The tactics, techniques, and procedures (TTPs) • The Indicators of Compromise (IoCs) • The hacking concepts and hacker classes • Ethical hacking concepts and scope • The different phases of hacking cycle • The ethical hacking concepts and its scope • The skills of an ethical hacker • Various ethical hacking tools	
3.	Information Security Threats and Vulnerabilities	3
	• The threat and threat sources • Malware and components of malware • The common techniques attackers use to distribute malware on the web • Different types of malware and malware countermeasures • The vulnerability and vulnerability classification • The vulnerability research • Vulnerability assessment and the types of vulnerability assessment • Vulnerability scoring systems and vulnerability management life cycle • The vulnerability assessment tools and vulnerability exploitation	
4.	Password Cracking Techniques and Countermeasures	2
	• The password cracking and password complexity • The Microsoft authentication mechanisms	
	• Various types of password attacks • Use different password cracking tools • Adopt countermeasures against password cracking attacks	
5.	Social Engineering Techniques and Countermeasures	2
	• Social engineering concepts • Perform social engineering using various techniques • Insider threats • Identity theft • Apply social engineering countermeasures • Apply knowledge of insider threats and identity theft countermeasures	

6.	Network Level Attacks and Countermeasures	4
	• Packet sniffing and types of sniffing • Different types of sniffing techniques • Use different sniffing tools • Apply various sniffing countermeasures • Different types of DoS and DDoS attacks • Use different DoS/DDoS attack tools • Apply knowledge of DoS/DDoS attack countermeasures • Implement different DoS/DDoS protection tools • The session hijacking process and types of session hijacking • Use different session hijacking tools • Apply knowledge of session-hijacking countermeasures	
7.	Web Application Attacks and Countermeasures	4
	• Web-server operations and security issues • Various web-server attacks and web-server attack tools • Adopt countermeasures against web-server attacks • Different web-server security tools • The web-application architecture and vulnerability stack • Various web-application threats and attacks • Use different web-application attack tools • Adopt countermeasures against web-application attacks • Use different web-application security tools • Different types of SQL injection attacks • Use different SQL injection tools • Adopt countermeasures against SQL injection attacks • Use different detection tools for SQL injection	
8.	Wireless Attacks and Countermeasures	3
	• Wireless terminology • Different wireless encryption algorithms • Wireless network-specific attack techniques • Use different wireless attack tools • Bluetooth attack techniques • Apply wireless attack countermeasures • Use different wireless security tools	
9.	Mobile Attacks and Countermeasures	2

	• Anatomy of mobile attacks • Mobile platform attack vectors and vulnerabilities • The importance of mobile device management (MDM) • Adopt various mobile security countermeasures • Use various mobile security tools	
10.	IoT and OT Attacks and Countermeasures	2
	• IoT concepts • Different IoT threats and attacks • Use different IoT attack tools • Apply countermeasures to protect devices from IoT attacks • Use different IoT security tools • OT concepts • Different OT threats and attacks • Use different OT attack tools • Apply countermeasures to protect industrial facilities from OT attacks • Use different OT security tools	
11.	Cloud Computing Threats and Countermeasures	2
	• Cloud computing concepts • Container technology • Cloud computing threats • Cloud computing attacks • Apply cloud computing security measures • Use various cloud computing security tools	
12.	Penetration Testing Fundamentals	2
	• Penetration testing and its benefits • Types and phases of penetration testing • Penetration testing methodologies • Various guidelines and recommendations for penetration testing • Various risks associated with penetration testing	

Recommended Study Material:

❖ **Text books:**

1. Michael T. Simpson, Kent Backman, James Corley — Hands-On Ethical Hacking and Network Defense, 3rd edition, Cengage, 2019.
2. Michael E. Whitman, Herbert J. Mattord, Principles of Information Security, 6th Edition, Cengage Learning, 2017
3. Chwan-Hwa (John) Wu, J. David Irwin, Introduction to Computer Networks and Cybersecurity, 1st Edition, CRC Press, 2017

❖ **Reference Books:**

1. Hacking Exposed: Network Security Secrets & Solutions, by Stuart McClure, Joel Scambray, George Kurtz, 7th Edition, McGraw-Hill, 2012
2. Kevin Mitnick, William L. Simon, The Art of Intrusion: The Real Stories Behind the Exploits of Hackers, Intruders & Deceivers, 1st Edition, Wiley, 2005
3. Joshua S. Wright, Wireless Networks Security: Theory and Practice, 1st Edition, CRC Press, 2016

Charotar University of Science and Technology (CHARUSAT)
Faculty of Technology and Engineering (FTE)

Subject: Network Defence Essentials (ITUE303) (University Elective - I)

Semester: 5th

Teaching Scheme:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/Week	2	1	-	4	3
Marks	50	50	-	100	

Course Pre-requisites:

Basic understanding of computer networks, operating systems, and internet technologies. Familiarity with concepts such as IP addressing, OSI/TCP-IP models, and basic cybersecurity terminology will be beneficial. Knowledge of fundamental programming concepts and system administration basics is desirable.

Course Outcomes:

By the end of the course, students will be able to:

CO1:Comprehend and differentiate the fundamental concepts of networking and network defense across different layers of the OSI model, including security goals, protocols, and defense mechanisms.

CO2:Analyze and evaluate different network defense controls such as administrative, physical, and technical mechanisms to secure information systems.

CO3:Apply security measures for modern computing environments, including virtualization, cloud computing, wireless networks, mobile devices, and IoT systems.

CO4:Demonstrate the ability to use cryptographic techniques, data security mechanisms, and network traffic monitoring tools to detect, prevent, and respond to security threats.

Bridge Topics/Self-Study/Revisit:

1. Basics of Computer Networks and Communication Systems
2. OSI and TCP/IP Reference Models
3. IP Addressing, Subnetting, and Routing Fundamentals
4. Network Devices: Switches, Routers, Firewalls, Access Points
5. Introduction to Cybersecurity Concepts and Threat Landscape
6. Identification, Authentication, and Authorization Basics
7. Fundamentals of Cryptography and Data Protection
8. Wireless Network Concepts and Security Basics
9. Cloud Computing and Virtualization Fundamentals
10. Mobile Device and IoT Security Concepts
11. Network Monitoring and Packet Analysis Basics
12. Security Policies, Compliance, and Risk Management

Syllabus:

Unit	Topics	Hours
1.	Information Security Fundamentals	3
	• The goals of network defense • The information assurance principles • The types of network defense approaches • The benefits and challenges of network defense • The different types of network security controls • The different network security protocols	
2.	Ethical Hacking Fundamentals	2
	• The terminology, principles, and models of access control • Describe identity and access management (IAM) • Understand user access management • The different types of authentication • The different types of authorization • Understand user accounting	
3.	Information Security Threats and Vulnerabilities	2
	• Various regulatory frameworks, laws, and acts • Understand why organizations need compliance • The need for and characteristics of good security policy • How to design and develop security policies • The different types of security policies • The different types of security and awareness training	
4.	Password Cracking Techniques and Countermeasures	2
	• The importance of physical security • The physical security attack vectors • The various types of physical security controls • The importance of workplace security • Physical security policy • The various environmental controls	
5.	Social Engineering Techniques and Countermeasures	4
	• Network segmentation and its types • The different types of firewalls and their roles • The different types of IDS/IPS and their roles • The different types of honeypots • The different types of proxy servers and their benefits • The fundamentals of virtual private networks (VPNs) and their importance in network security • Security incident and event management (SIEM) • User behavior analytics (UBA) • Apply various antivirus/anti-malware software	
6.	Network Level Attacks and Countermeasures	3
	• Virtualization, its components, and virtualization enablers • OS virtualization security and concerns	

	• The best practices for OS virtualization security • Cloud computing and its benefits • Different types of cloud computing services • Cloud deployment models • Importance of cloud security and its best practices	
7.	Web Application Attacks and Countermeasures	2
	• Wireless terminology, wireless networks, and wireless standards • The wireless network topologies and classification of wireless networks • The components of a wireless network • The wireless network encryption mechanisms • The different types of wireless network authentication methods • Wireless network security measures and wireless security tools	
8.	Wireless Attacks and Countermeasures	2
	• The various mobile device connection methods • The concepts of mobile device management • The common mobile use approaches in enterprises • The security risk and guidelines associated with enterprise mobile usage policies • Enterprise-level mobile security management solutions • The general security guidelines and best practices for mobile platforms	
9.	Mobile Attacks and Countermeasures	3
	• The IoT and why organizations opt for IoT-enabled environments • The IoT application areas and IoT devices • The IoT architecture and IoT communication models • The security in IoT-enabled environments and stack-wise IoT security principles • The security considerations of the IoT framework • IoT device management • The best practices and tools for IoT security	
10.	IoT and OT Attacks and Countermeasures	3
	• Cryptographic techniques • The different encryption algorithms • The different hashing algorithms • Use different cryptography tools and hash calculators • Public key infrastructure (PKI) • Digital signatures and digital certificates	
11.	Cloud Computing Threats and Countermeasures	2
	• Data security and its importance • The different data security technologies • The various security controls for data encryption • Use different disk encryption, file encryption, and removable-media encryption tools • Methods and tools for data backup and retention • Data loss prevention (DLP) and DLP solutions	
12.	Penetration Testing Fundamentals	2

	• The need for and advantages of network traffic monitoring • The network traffic signatures • The categories of suspicious traffic signatures • The attack signature analysis techniques • Network monitoring for suspicious traffic • The various network monitoring tools	
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Recommended Study Material:

❖ **Text books:**

1. Douglas R. Stinson, Cryptography: Theory and Practice, 4th Edition - CRC Press (2018)
2. William Stallings, Network Security Essentials: Applications and Standards, 7th Edition - Pearson (2017)

❖ **Reference Books:**

1. Behrouz A. Forouzan, Cryptography and Network Security, 4th Edition - McGraw-Hill (2017)
2. Charlie Kaufman, Radia Perlman, Mike Speciner, Network Security: Private Communication in a Public World, 2nd Edition - Prentice Hall (2002)
3. Ross Anderson, Security Engineering: A Guide to Building Dependable Distributed Systems, 3rd Edition - Wiley (2020)